

B.T. Franklin

Phoenix, Arizona

Applied AI Technical Lead / Software Architect

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Professional Summary

Applied AI technical lead and software architect with 20+ years of experience across enterprise software, developer tooling, and product-facing systems.

- Focused on generative AI, large language models (LLMs), agentic systems, multimodal AI integrations, and AI product architecture.
- Experienced in moving emerging AI capabilities into useful engineering workflows and product experiences.
- Combines full-stack engineering depth with product thinking, UX sensibility, technical leadership, and published computational creativity research.

Core Expertise

- Applied AI and Large Language Model (LLM) Systems
- Agentic Workflow Design and Evaluation
- Model Context Protocol (MCP) Server Architecture
- Retrieval-Augmented Generation (RAG), Embeddings, and Semantic Matching
- Multimodal AI API Integration
- AI Product Architecture
- Developer Experience and Platform Tooling
- Full-Stack Product Engineering
- Technical Leadership

AI Platform Experience

- Built with OpenAI GPT models from GPT-3 through current models, OpenAI image generation models, OpenAI embedding models, OpenAI speech-to-text models, and OpenAI interactive voice models across professional and serious side projects.
- Built with Anthropic Claude models, ElevenLabs speech generation models, and production-oriented LLM API workflows.
- Designed agentic workflows involving tool use, planning loops, multi-agent systems, background processing agents, self-correcting guardrail loops, and custom headless processing loops.

- Built RAG, embedding, vector-search, semantic matching, structured extraction, and document-processing systems using custom pipelines and vector stores.
- Applied AI quality and evaluation practices including rubric grading, human review, tracing, observability, guardrail-style parsing checks, skill/workflow evaluation, automation-oriented review loops, and practical output quality checks.

Domain Experience

- Deepest domain experience is in pricing and revenue optimization, including pricing workflows, price management systems, formula-driven price calculations, analytics, and customer-facing explanation of pricing outputs.
- Historical domain experience in Salesforce and GTM systems, including Salesforce UI/UX, Lightning Web Components, enterprise implementation strategy, and configuration migration workflows.
- Strong working knowledge of GitHub and repository intelligence, including repository architecture analysis, contributor expertise modeling, development-history analysis, agent-legible repositories, and GitHub-integrated AI product workflows.
- Practical recruiting-domain experience from building Hiredar, including resume ingestion, candidate profile extraction, explainable candidate matching, recruiter-facing workflows, and job-to-candidate semantic comparison.
- Earlier medical imaging software experience, including refactoring product systems that expanded clinical applicability and strengthened product value.

Architecture and Technical Leadership Evidence

- Served as chief architect, design owner, proposal author, and implementation driver for end-to-end enterprise systems across public APIs, analytics integrations, AI workflows, developer tooling, and customer-facing product surfaces.
- Architected a data access API that exposed a public server-side surface for CRUD-style data operations against Zilliant's legacy microservice stack, enabling integrations that did not require users to work through the web UI.
- Architected and led implementation of a full AWS QuickSight and Amazon Q analytics integration, embedding generative AI-assisted analytics and on-demand visualization workflows into the product UI.
- Architected and led Zilliant Developer Experience (ZDX), a CLI that migrated complex solution configurations between deployed environments and reduced a previously multi-day process to a few hours.
- Led technical strategy across roughly six engineering teams without direct people-management authority, serving as a bridge between siloed teams, product management, engineering leadership, and implementers.

- Mentored engineers through domain-specific skill transfer, design explanation, and implementation guidance, including sustained mentorship for junior engineers and situational mentoring for senior engineers working outside their strongest domains.
- Created proposals, design documents, Confluence guidance, proof-of-concept implementations, and reference implementations so teams could align around technical direction after initial design discussions.
- Balanced product requirements, engineering constraints, trade-offs, and ambiguous implementation paths by translating between product managers and engineering teams until the group reached a design decision with understood constraints.
- Strongest architecture areas include framework design, API design, platform design, developer tooling, UI/UX architecture, AI workflow design, and agent-assisted developer workflow design.
- Created a public AI agent skills repository and brought reusable skills back into team workflows to accelerate development practices such as code review and agent-guided engineering tasks.

Differentiators

- Long-running AI orientation that predates the current LLM wave, including graduate-level AI coursework in the 1990s, years of independent computational creativity research, and practical exposure to expert-system, game-intelligence, generative-art, and modern LLM/agent AI paradigms.
- Builder by default: repeatedly creates working products, frameworks, libraries, games, research systems, and open-source tools outside formal job requirements, using code as the primary way to explore ideas and turn technical taste into usable systems.
- Combines enterprise architecture judgment from long Zilliant tenure with founder-style product execution from solo SaaS builds, public open-source tooling, marketing experiments, deployment work, and go-to-market attempts.
- Brings a broad AI perspective spanning agent workflows, tool use, MCP, structured prompts, RAG, semantic matching, computational creativity, human-facing AI product UX, developer workflow acceleration, and evidence-grounded automation.
- Comfortable operating in ambiguous spaces where product goals, technical constraints, user experience, and evolving AI capabilities have to be reconciled into a design that engineers can actually build.

Professional Experience

Zilliant, Inc.

AI Tech Lead / Staff Software Engineer

April 2025 - July 2026

- Served as organization-wide AI technical leader, guiding engineering teams on adoption of large language models (LLMs), agents, evaluation practices, and modern AI development patterns aligned with business goals.
- Architected the company's first Model Context Protocol (MCP) server and directed implementation by a third-party contractor, connecting the classic price management stack and formula evaluation service to agent workflows.
- Designed MCP tools that allow agents to retrieve calculated prices and explain the formula reasoning behind start, target, and floor price values shown in customer-facing pricing experiences.
- Shipped the MCP server work for existing customers on Zilliant's older stack, supporting modernization, customer retention, and license-expansion opportunities without relying on unverified revenue attribution.
- Led cross-team technical strategy for AI adoption, translating emerging AI capabilities into practical engineering workflows and product opportunities across product management, engineering leadership, and implementation teams.
- Drove broad adoption of AI tooling across engineering, including Codex workflow training, reusable code-review skills, and agent-assisted development patterns that helped move PR review and code-production workflows toward faster turnaround.
- Provided ongoing mentorship to engineers at all levels of seniority.

Software Architect

August 2022 - March 2025

- Designed and implemented a custom generative AI pipeline that enabled real-time integration of diverse pricing insights into customer-facing analytics experiences.
- Led integration of AWS QuickSight and Amazon Q, embedding analytics visualization and generative AI-assisted question answering into the product UI to enhance customer decision support.
- Shipped the AWS QuickSight and Amazon Q integration as a customer-facing analytics capability.
- Served as chief architect for a data access API that created a server-side integration surface for CRUD-style operations against Zilliant's data and legacy microservice stack.
- Designed and launched the Zilliant Developer Experience (ZDX) CLI, streamlining internal and external developer workflows and reducing complex configuration migration work from a multi-day process to a few hours.
- Helped shape architectural direction across AI, analytics, and developer enablement initiatives.

Staff Software Engineer

February 2020 - August 2022

- Directed UI/UX strategy for enterprise Salesforce implementations, improving usability and aligning product experiences with business objectives.
- Developed strategic proof-of-concept solutions that influenced company-wide technology direction and adoption decisions.
- Facilitated collaboration across globally distributed teams to align implementation choices with broader technical strategy.

Senior Software Engineer

January 2009 - September 2019

- Championed adoption of Salesforce Lightning Web Components, contributing to major UI modernization and improved application performance.
- Led development efforts on SalesMax and mentored interns through successful product delivery, increasing usability and competitiveness.
- Contributed across full-stack engineering, front-end architecture, and product modernization initiatives over a decade of platform evolution.

Selected Founder-Led AI Products

RepoZest - *Founder / Product Architect, 2026*

- Built and publicly deployed a solo GitHub App-backed repository intelligence SaaS, later sunset and open-sourced as an archived reference project after market validation.
- Designed a Django monolith with web, worker, and beat services; GitHub App authentication and installation scope; repository indexing; analytics derivation; AI artifact generation; billing; operator pages; and Sevalla deployment infrastructure.
- Built deterministic repository extraction over branch-aware snapshots, file trees, representative source excerpts, commit history, contributor aliases, pull requests, review context, path-level touch facts, and contributor-path rollups.
- Designed an intelligence pipeline that derived architecture areas first, mapped paths to semantic system areas, and used those architecture mappings to ground contributor expertise inference.
- Produced architecture reports, expertise reports, repository-specific expert-routing suggestions, hotspot and maintenance-pressure signals, and interactive expertise-routing answers from published expertise graph state.
- Designed the product around evidence-backed repository understanding: architecture visibility, current-team-first expert discovery, knowledge concentration, hotspots, ownership fragility, and coordination risk.
- Implemented provider-neutral billing architecture with Stripe-first hosted billing, GitHub Marketplace as a later adapter direction, repository entitlement enforcement, trial policy, and subscription normalization.

- Created sanitized deployment infrastructure with Terraform, GitHub Actions workflows, production-shaped manifests, helper scripts, and operator runbooks for Sevala deployment.
- Owned end-to-end product execution, including opportunity definition, product strategy, system architecture, UX, AI pipeline design, GitHub integration model, billing model, deployment model, and go-to-market positioning.

Hiredar - *Founder / Product Architect, 2025*

- Built and deployed a complete solo AI recruiting SaaS in roughly two months, using AI tooling to accelerate product, engineering, UI, deployment, and go-to-market execution.
- Conceived, architected, and launched an AI recruiting platform focused on resume parsing, structured candidate extraction, and AI-assisted candidate-to-role matching.
- Built a self-service recruiter workflow with sign-up, authentication, email verification, recruiter dashboard, job opening creation, bulk resume uploads, candidate profiles, match lists, candidate detail views, and shortlist export.
- Built a Django/Celery background processing pipeline that ingested resume documents, extracted structured candidate data with OpenAI APIs, generated embeddings, and stored matching vectors in Pinecone.
- Designed a custom RAG and semantic matching system that compared candidate-profile sections against job-description sections across skills, experience, career direction, qualifications, and holistic fit.
- Implemented a multi-lens candidate matching engine with holistic, skills, relevant experience, wildcard, and qualifications perspectives backed by Pinecone vector namespaces for candidate profiles and job openings.
- Built credit-based monetization with free starting credits, Stripe Checkout for credit purchases, recruiter credit balances, premium-action credit deduction, and no-subscription/pay-as-you-go pricing.
- Built HTMX-driven progress and polling UI for long-running background jobs, including task metadata, status fragments, and per-owner async job visibility.
- Deployed the application with a hosted database, production-oriented settings, Celery workers, queued email handling, S3-style storage configuration, and Sevala-oriented production deployment concerns.
- Created and ran go-to-market assets and campaign work, including positioning, marketing strategy, landing-page messaging, demo/video concepts, and recruiter-focused direct-response campaign materials.
- Implemented guardrail-style checks and corrective loops around resume parsing and candidate extraction quality.
- Led the full product lifecycle independently, including product strategy, application architecture, full-stack implementation, UX design, deployment, pricing model definition, and live-market validation.

Selected Open Source AI Work

- Built and published open source AI and developer tooling spanning agent orchestration, AI grounding, prompt engineering, and engineering automation.
- Created contract4agents, a typed declarative contract language and toolchain for defining AI agents with callable interfaces, context requirements, capabilities, policies, guards, assertions, eval packs, monitor rules, and provider-neutral manifests.
- Created agentic-django, a Django-native integration for building agentic apps with the OpenAI Agents SDK, persistent sessions and runs, tool calling, background tasks, and HTMX/REST status polling.
- Created agent-simulator, a Django/Channels application for simulating realistic agent conversations against configurable mock MCP tools using real provider APIs before building the actual MCP server.
- Created and maintains a public skills repository with reusable AI agent skills for Django, OpenAI, web UI, Python packaging, structured LLM workflows, code review, production-readiness review, and agent-legible repository work.
- Created compendiumscribe, an OpenAI Agents SDK research workflow that decomposes topics into planning, web research, verification, and synthesis stages; validates structured Pydantic outputs; maintains a citation ledger; and renders sourced compendiums as Markdown, XML, HTML, or PDF.
- Created promptdown, a Python package for expressing structured LLM prompts in Markdown files, separating prompt assets from code and supporting reusable prompt engineering workflows.
- Created release-notes-scribe, a GitHub Action that uses the OpenAI Responses API to generate draft GitHub release notes from tag-to-tag commits and diffs.
- Created pywebview-htmx, a Python desktop UI library that brings HTMX-style declarative interactions to PyWebview apps through Python-bound UI attributes, target swaps, loading states, concurrency controls, and lifecycle events.
- Other notable projects include wordsmith-engine, a composable deterministic-friendly text generation toolkit for Python, plus earlier generative creativity projects connected to published computational creativity work.

Additional Experience

- **Senior Salesforce Engineer, Camping World, Inc.** (Sep 2019 - Feb 2020): Rewrote a complex store location management application using a more extensible architecture and modern development practices.
- **Senior Software Engineer, Confirma, Inc.** (2006 - 2008): Refactored medical imaging software, expanding clinical applicability and strengthening product value.
- **Senior Software Engineer, Teranode Corporation** (2005 - 2006): Built intuitive data-management interfaces that improved user productivity and accuracy.

- **Software Engineer, Digital Harbor, Inc.** (2000 - 2005): Developed data-driven user interfaces supporting strategic analysis and decision-making.

Research and Publications

- Franklin, B.T. "Spooklight: A Generative Approach to AI-Driven Storytelling," *Proceedings of Generative Art Conference*, vol. 27, Dec. 2024, pp. 50-59.
- Franklin, B.T. "Resonant Element: An Application for the Continuous Generation of Pop-Inspired Music," *Proceedings of the 3rd Computer Simulation of Musical Creativity Conference*, Aug. 2018.
- Franklin, B.T. "SongSkill: A System for Continuous, Emotionally-Adaptive Music Generation," *Proceedings of Generative Art Conference*, vol. 19, Dec. 2016, pp. 124-37.

Education

Transylvania University, Lexington, KY

BA in Computer Science, 1997

Minor in Psychology

- Graduated with Honors
- Recipient of the Computer Science Award
- First student in school's history to complete three Senior Honors projects

Skills

Programming Languages: Python, Java, JavaScript/TypeScript, Apex, Swift

AI Models and APIs: OpenAI GPT Models, OpenAI Image Generation, OpenAI Embeddings, OpenAI Speech-to-Text, OpenAI Interactive Voice, Anthropic Claude, ElevenLabs Speech Generation

Technologies: Large Language Model (LLM) Applications, Agentic Systems, Model Context Protocol (MCP), RAG, Embeddings, Vector Stores, Semantic Search, Pinecone, Structured Extraction, REST APIs, Django, Celery, HTMX, Pydantic, PydanticAI, OpenAI Agents SDK, Codex, Docker, PostgreSQL, SQLite, AWS Services, AWS QuickSight, Amazon Q, CI/CD, GitHub Actions, Terraform, React, Tailwind CSS, Salesforce Lightning, Lightning Web Components, Visual Studio Code, oclif

Methodologies: AI Development, Prompt Engineering, Tool Use, Planning Loops, Multi-Agent Systems, Guardrail Design, Rubric Grading, Human Review, Tracing, Observability, Large Language Model (LLM) Evaluation, Agent Evaluation, Framework Design, API Design, Platform Design, Reference Implementations, Technical Strategy, Developer Workflow Optimization, Agile

Other Skills: Software Architecture, AI Product Architecture, Full-Stack Development, UI/UX Design, Developer Experience, Cross-Functional Leadership, Mentorship, Git, Perforce, Jira, Confluence, HTML/CSS